Hypothesis: The positive damping would suppress oscillations while negative damping would amplify it, nonlinear Van der Pol feedback would produce a stable limit cycle. The graph shown on the right side supported the hypothesis.

α	Oscillation period
0.5	0.800 s
5	0.877 s
20	1.494 s

The oscillation period increased from 0.800 s to 1.494 s (approximately 87%) as alpha increased from 0.5 to 20, while the steady-state amplitude remained approximately 2.

Conclusion: Mathematical models can reproduce complex biological behaviors using relatively simple equations, but the outputs depend strongly on parameter choices before being considered realistic.

